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LECTURES 06b–08a • CLUSTERING • HW 3a/3b • CONCEPTS, THEORY, PRINCIPLES, LIGHT CODE INTERPRETATION

CLUSTERING: PURPOSE + REPRESENTATION

- A clustering is a proposal under chosen observations, features, transforms, scales, distance, method, and settings. Evaluate it for its intended use; an algorithmic result is not proof of natural populations or a decision.
- Geometry comes from representation. Nominal integer codes invent order/gaps; one-hot makes category mismatches equally distant. Standardization prevents large numerical units from dominating but does not establish equal substantive importance.
- HW3B: six annual spending features; use $\log(1+x)$ to compress extreme spending, then `StandardScaler`. Customer ID is not a feature. Describe groups using original-unit medians and counts. A 2-D Fresh/Grocery plot cannot show all six dimensions.
- Validation questions: Does structure match the purpose? What does another group add? Does it persist across settings/features/starts/samples? Could a justified one-population reference produce it? Does it relate to external outcomes or another period?

K-MEANS: OBJECTIVE + FIT

- Request K; initialize centers; assign each row to nearest center; replace each center with assigned mean; repeat until movement $<$ `tol` or `max_iter`. Each step reduces/preserves the within-cluster sum of squares (inertia).

$$W = \sum_i \|x_i - \mu_{c(i)}\|^2 \text{ (inertia/SSE)}$$

- `k-means++` favors initial rows far from chosen centers; `random` samples rows. Later centers are means and need not be observed. `n_init` runs complete fits and keeps lowest *final* inertia, not best initial inertia or most common partition. `random_state` reproduces randomized choices.
- Course-version defaults: $K=8$, `max_iter=300`, `tol=.0001`, algorithm Lloyd. Learned: `labels_`, `cluster_centers_`, `inertia_`, `n_iter_`. `predict` assigns nearest fitted centroid; `transform` returns distances to centers. Reuse fitted preprocessing.
- Example $\{1,2,8,9\}$, start centers 1,2: start inertia $6^2+7^2=85$; settled centers 1.5,8.5; inertia $4(0.5^2)=1$. **Limits:** forces all rows into groups; compact/roughly spherical bias; outliers pull means; labels/colors arbitrary; $K=3$ may split one broad population; near-equal inertia may encode different operational stories.

CHOOSING K

- Inertia cannot select K: it never increases when centers are added. Elbow seeks diminishing improvement and can be ambiguous.
- Silhouette: $a(i)$=mean distance to own cluster; $b(i)$=smallest mean distance to another; $s(i) = [b-a]/\max(a,b)$. Near 1 separated; near 0 overlap; negative closer to another cluster; undefined at $K=1$. Mean silhouette favors compact separated groups and may undervalue curved structure.
- Gap: compares observed inertia to reference (no-group) simulations; choose first k where improvement falls within simulation noise.

$$\text{Gap}(k) = B^{-1}\sum_b \log W^*_{kb} - \log W_k;$$

$$s_k = \text{SD}(\log W^*_{kb})\sqrt{1+1/B};$$

$$\text{Choose first } k \text{ with } \text{Gap}(k) \geq \text{Gap}(k+1) - s_{k+1}$$

- Depends on no-group reference; bars are simulation allowances, not CIs; not a p-value/proof of true K.
- Close diagnostics ⇒ prefer simpler K unless added group is stable, interpretable, and changes useful action. Inspect counts, profiles, original records, and another period.

HIERARCHICAL CLUSTERING

- Agglomerative: one cluster/row; repeatedly merge closest current pair under linkage; record children, height, size; continue to one. Greedy/irreversible; early merges cannot be undone; a cut need not globally optimize its cluster count.
- Single=min cross-pair (connected/elongated; chaining). Complete=max (keeps all members relatively close). Average=mean cross-pair. Ward=smallest increase in within-cluster SSE (Euclidean; compact).
- Example $d(A,C)=3.16$, $d(B,C)=2.24$: single 2.24, average 2.70, complete 3.16. Merge height is linkage value, not every within-cluster pair distance.
- Dendrogram: leaves=observations; vertical joins=linkage values; horizontal spacing is layout only. Follow branches. Horizontal cut gives a flat partition. Cuts of one tree are nested; separate K-means fits need not be.
- SciPy `linkage(X,method,metric)` returns $(n-1)\times 4$: child 1, child 2, height, size. IDs $0\ldots n-1$ are rows; new cluster IDs start at n. `dendrogram` only draws existing history.
- `fcluster(...,criterion="distance")` uses t as height. `maxclust` yields at most t groups; ties may yield fewer. `cut_tree(...,n_clusters=k)` follows merge order to request a count. Labels are arbitrary.
- Ordinary agglomerative clustering has no built-in `predict`/decision boundary for future rows; add a rule or refit/reassess.

DBSCAN

- `eps`=radius; points exactly at radius count. `min_samples`=required neighborhood count including self.
- Core: count $\geq$ minimum. Border: non-core within `eps` of a core. Noise: neither; sklearn label -1. Clusters grow through connected cores, then absorb borders. **Border is not a bridge:** it joins but cannot extend to its own neighbors.
- Larger `eps`: more neighbors, less noise, possible merging. Smaller: fragmentation/more noise. Larger `min_samples`: harder core status/more noise. Scaling/metric change neighborhoods.
- Strength: irregular/curved connected regions and explicit noise; endpoints may be farther than `eps` via core chains. Weakness: one global density rule struggles with unequal densities. Seek a stable range, not one attractive image.
- Learned: `labels_`, `core_sample_indices_`. Noise may be error, rare legitimate case, or setting artifact. No built-in sklearn new-row prediction rule. **HW3B radius lesson:** with the same six-feature X and `min_samples=8`, $\epsilon = .8$ gave 2 groups/147 assigned/293 noise; $\epsilon = 1.2$ gave 1 group/355 assigned/85 noise. More coverage can erase distinct profiles.

GAUSSIAN MIXTURE MODELS

$$p(x) = \sum_{g=1}^{G} \pi_g \mathcal{N}(x \mid \mu_g, \Sigma_g), \pi_g \geq 0, \sum_g \pi_g = 1$$

$$\text{Posterior } P(g \mid x) = \pi_g \mathcal{N}_g(x) / \sum_h \pi_h \mathcal{N}_h(x)$$

- EM: initialize means/covariances/weights; E-step membership probabilities; M-step probability-weighted parameter updates; repeat until likelihood improvement is small/limit. Multiple starts retain highest likelihood. Check `converged_`, `n_iter_`.
- `predict_proba` preserves soft membership; `predict` gives argmax label. A .516/.484 label is ambiguous. High relative membership can occur where every component has low absolute density.
- Covariance: full=own tilt/shape/size; tied=one shared covariance; diag=own axis-aligned variances; spherical=one variance/component. Full has $p(p+1)/2$ covariance values/component; diag p; spherical 1.
- Extra components may approximate skew/non-Gaussian shape, not populations. Ellipses show component spread/equal density, not hard boundaries or mean CIs. Weights need not equal hard-label fractions. **HW3B ambiguity:** customer 275 probabilities (.306,.261,.432) show a mixed assignment hidden by hard label 2; customer 50 was decisive (.999,.001,.000).

METHOD COMPARISON

- K-means: compact center-based groups; requested K; hard labels; future nearest-center assignment; retains lowest inertia.
- Hierarchy: nested resolutions/merge history; linkage and cut determine geometry; deterministic absent ties; future assignment requires extra rule.
- DBSCAN: irregular density-connected regions + noise; number of groups is output; `eps`/`min_samples` critical; does not assign all rows.
- GMM: overlapping elliptical distributions + uncertainty; requested components; soft probabilities; retains highest likelihood.
- All depend on representation. Agreement across methods is not independent proof; disagreement should be explained through each method's rule.

CLUSTER VALIDATION + INFERENCE

- Internal geometry (inertia/silhouette/gap) is not truth/usefulness. Inspect members, profiles, added distinctions, settings/features/starts/samples, outside outcomes, another period, and domain knowledge.
- ARI compares pairwise co-membership while ignoring label numbers: 1 identical; about 0 chance agreement; negative below chance. Subsampling (e.g., 80%) checks sensitivity within observed data, not independent replication. Initialization and observation stability answer different questions. **Double dipping:** clustering on x, then ordinary t-testing group means on x, reuses the same information; K-means constructed separation. Tiny naive p can arise from one Gaussian population. Assigning holdout rows using the same features can still create differences by construction; shuffling final labels omits selection.
- Valid reference comparison: predefine features/statistic/null; simulate full datasets under justified reference; repeat all preprocessing, clustering, and count/feature search; compare observed statistic. Example $W_{K=2}/W_{K=1}$, smaller=stronger split.
- Monte Carlo tail: $(1+\#\{T_b \text{ as/extremer}\})/(B+1)$. Evidence against one Gaussian does not prove exactly 2 populations; skew/heavy tails may also reject it. Formal test needs calibrated false-positive behavior.
- If many methods/settings were tried and strongest reported, simulate the search or use fresh evaluation data. Multiple geometry scores are not independent confirmation.

CODE + OUTPUT RECOGNITION

- `StandardScaler().fit(Xtrain)` learns means/SDs; `transform` reuses them. `fit_transform` learns and applies to initial data.
- `KMeans(...).fit(X)`: inspect underscore attributes; `predict(newX)` uses fixed centers.
- `linkage(X,...)` returns merge array, not estimator. `cut_tree`/`fcluster` cut the existing tree; they do not refit.
- `DBSCAN(...).fit(X)`: cluster count excludes -1; count noise separately.
- `GaussianMixture(...).fit(X)`; `predict_proba` rows sum to 1; `predict` chooses maximum.
- Be able to infer shapes: n rows, p features; K-means centers $K\times p$; GMM probability output $n\times G$; linkage $(n-1)\times 4$.

PART A QUESTION CHECKLIST

- What features/scale/reference did the method use? What setting changes the evidence versus only the display/selection?
- What exact rule produced the assignment? What does the output establish, and what does it not establish?
- If code is shown: constructor settings have no trailing underscore; learned attributes usually do. Fit learns; transform/predict applies.
- If a plot is shown: identify original dimensions omitted, arbitrary labels/colors, and whether the plot is geometry, fit, or evidence.

HIGH-YIELD CLUSTER TRAPS

- Lower inertia with larger K is automatic; it does not validate K.
- K-means centers and GMM means are not necessarily observed rows.
- Standardization changes assignments; it does not prove equal feature importance.
- Same plotted location in two features does not imply proximity in all fitted features.
- Dendrogram horizontal spacing is not distance; merge height depends on linkage.
- Single linkage can chain; complete can isolate small extreme groups; Ward favors compact groups.
- DBSCAN border joins but cannot propagate; noise is not automatically bad data.
- Larger `eps` can merge groups even if noise count changes little.
- GMM hard labels hide uncertainty; strong relative membership is not absolute typicality.
- Stable/separated $\neq$ causal, real, or useful. Profiles describe fitted members; they do not prove customer types or business causes.

HW 3A/3B: WHAT TO REMEMBER

- Clustering quality is purpose-dependent. Technical cohesion/separation/stability are useful supporting evidence, not a universal definition of a good clustering.
- Benchmark class labels are only one possible structure. A defensible clustering can differ from known labels if it captures another useful relationship.
- K-means 3-group wholesale profiles: report counts and medians, but avoid inferring business type. Group overlap in a 2-D plot is expected when fitting six features.
- Complete-linkage three-group summary placed 402/440 in one group and only 5 in the smallest: a valid cut can still be too broad/imbalanced for the intended account-review purpose.
- DBSCAN settings trade assigned coverage against distinct groups; noise customers still warrant review rather than automatic exclusion.
- GMM probabilities can change a review by exposing mixed profiles instead of presenting one categorical label.

QUICK METHOD TABLE

K-means	nearest center; SSE; hard; all assigned
Hierarchy	linkage merges; nested cuts; all assigned
DBSCAN	core connectivity; noise; count emerges
GMM	Gaussian density; soft probabilities; all covered

FAST CALCULATIONS

- Euclidean $d(x,y) = \sqrt{\sum_j (x_j - y_j)^2}$; squared distance drives K-means.
- Silhouette example $a=2, b=5$: $(5-2)/5 = .60$.
- Linkage cross-distances 2 and 8: single=2; average=5; complete=8.
- DBSCAN: count self and boundary points at exactly `eps`.
- GMM probabilities per row and mixture weights each sum to 1.

SCOPE CONTROL

Not Part A scope: Lecture 05a embeddings are before the announced 06b start. Lecture 12a forecasting is after the instructor's stated Part A emphasis "through 11b." Use page 2 for anomaly detection and time-series fundamentals through 11b.

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LECTURES 09b–11b • ANOMALY DETECTION + TIME-SERIES FUNDAMENTALS • HW 4 + RELEVANT HW 5a AUDIT CONCEPTS

ANOMALY FOUNDATIONS

- A detector converts a chosen meaning of unusualness into a score. Reference defines unusual: mixed collection vs new readings against accepted healthy baseline (novelty). Check operating/ context comparability.
- Point anomaly: one row departs. Contextual: unusual only within condition/ regime. Collective: relationship/ joint pattern is unusual even when individual values are familiar.
- SCORE = method-specific unusualness; RULE = threshold/ rank/ budget; FLAG = selected for follow-up; FAULT/ OUTCOME = independently established condition. Do not equate them.
- Workflow: define target/ reference → features/ scale → fit → orient scores → rule based on cost/capacity → investigate records/ context/ sensors → validate with external outcomes/ new data.

LOF: FORMULAS + MEANING

- For point p, $N_k(p)$ excludes self. $r_k(o)$ is distance from receiving neighbor o to its kth-nearest other point.

$$\text{reach}_k(p,o) = \max\{d(p,o), r_k(o)\}$$

$$\text{lrd}_k(p) = \left[\frac{1}{|N_k(p)|}\sum_{o\in N_k(p)} \text{reach}_k(p,o)\right]^{-1}$$

$$\text{LOF}_k(p) = \frac{|N_k(p)|^{-1}\sum_{o\in N_k(p)} \text{lrd}_k(o)}{\text{lrd}_k(p)}$$

- Reachability is directional because receiving neighbor o supplies its radius. LOF near 1=comparable local support; > 1=lower density than neighbor average. Absolute sparsity alone need not be locally unusual.
- Values 0, 1, 3, 10, 50, 60, 80, $k = 2$: for 10, neighbors 3,1; reaches 7,9; lrd=.125; neighbor lrds .400,.333; LOF=2.933. For 80, reaches 20,30; lrd=.04; LOF=.917. At $k = 3$, 10 falls to .953 and wide group rises to about 2.718.
- k changes neighborhood evidence and can reverse rankings, so refit. Threshold changes only selection from fixed scores, so no refit. Scaling/features/metric also alter neighborhoods.
- sklearn fitted collection: `-negative_outlier_factor_` gives larger=unusual. `kneighbors()` returns ordinary distances; reachability is internal. New rows: fit `LocalOutlierFactor(novelty=True)` on reference, reuse scaler, score unseen rows only.

ISOLATION FOREST

- Each node randomly chooses a feature and split between that node's min/max; repeat. Short average path = easy isolation = more unusual after sample-size normalization. Endpoint/ large-gap rows often isolate quickly.
- Forest averages trees, often built from row samples. `n_estimators` = trees; `max_samples` = rows/tree; `random_state` = repeatability. More trees reduce dependence on one random sequence.
- sklearn `score_samples`: lower=more anomalous; negate for larger=unusual. Score is not fault probability.
- LOF asks "less locally supported than neighbors?" Isolation Forest asks "easy to isolate?" Thus value 10 leads LOF while endpoint 80 leads isolation.
- LOF distances usually require deliberate scaling. Isolation Forest's one-feature splits are invariant to positive linear unit changes, so scaling is generally unnecessary for that reason.

THRESHOLDS, BUDGETS, METRICS

- For larger=unusual scores, lowering threshold can only preserve/add flags; workload and recall cannot decrease; precision can rise or fall. Scores/ranking stay fixed. Do not share a numerical cutoff across detector scales.
- Fixed budget: rank and review top N; cutoff varies by collection. Fixed numerical threshold: workload varies. State tie rule; ties may select $> N$.
- `contamination` calibrates a training cutoff/expected fraction; in lecture examples it changes flags/`offset_`, not raw scores or which rows influence the fitted reference.

Metric Definition	Formula
TP = flag + outcome	**Precision** = $TP/(TP+FP)$
FP = flag + no outcome	**Recall** = $TP/(TP+FN)$
FN = no flag + outcome	**Specificity** = $TN/(TN+FP)$
TN = no flag + no outcome	**FPR** = $FP/(FP+TN)$
$\mathbf{F_1}$ **Score** = $2PR/(P+R)$	**Flag**=Pred(Yes) | **Outcome**=Actual(Yes)

- Reviewing only flags can estimate precision, but recall needs total positives or representative review of unflagged rows. Flags count workload, not faults.

WINE VALIDATION + HW4

- Lecture red-wine reference: 1,599 red/4,898 white; 11 chemistry features; quality excluded. 150 mg/L sulfur dioxide exceeds 99.4% of red but 63.1% of white values: pooling obscures context.
- Red example: standardized chemistry; LOF $k = 20$; Isolation Forest 300 trees, `max_samples=256`; budget 50; cutoffs about 1.506/.554; only 19 overlap. Detector choice changes investigations.
- Low rating ≤ 4: 63/1599. LOF top 50: TP=7, FP=43, FN=56, TN=1493; precision 14%; recall 11.1%. Review 100 adds 3 low-rated +47 others: precision 10%, recall 15.9%.
- Uniform random expected count = 50(63)/1599 = 1.97. Hypergeometric one-sided $p \approx .00275$ for $X \geq 7$: fixed LOF list has stronger low-rating association than random review. Not causation, every flag, completeness, optimality, or future generalization.
- HW4 white wine: 183/4898 low-rated; budget 150 random expectation 150(183)/4898 = 5.60. LOF found 22 (14.7% precision, 12.0% recall); Isolation Forest 23 (15.3%, 12.6%); only 52 shared, 98 method-only each.
- HW4 LOF $k = 20 \rightarrow 10$: 106 stayed, 44 entered, 44 left; low-rated found 22→19; precision 14.7%→12.7%. Same workload does not mean a stable list.

CONTAMINATION + MASKING

- Reference contamination: target-abnormal behavior influences fitted reference. Masking: similar abnormal rows support one another and receive less unusual scores.
- Demo with LOF $k = 20$: remote group 5→30 makes focal LOF 6.19→1.01 because all 20 neighbors can become remote peers. LOF rank 5/205→125/230; Isolation Forest rank 2/205→25/230.
- Changing cutoff/`contamination` does not cure masking because raw relationships remain. Use accepted-normal reference, regime-specific comparisons, provenance checks, temporal holdout, and reference-composition sensitivity; investigate groups of anomalies.
- New-data evaluation: freeze features, scaler, reference model, and budget/threshold; score separate collection before outcomes; then reveal outcomes and compare with same-size baseline. Keep related/repeated rows together. Scoring preserves reference; refitting updates it and can introduce masking.

TIME INDEX + MOVING AVERAGES

- Parse dates with explicit format; sort; require unique labels, no missing values, expected complete calendar (`freq="QS"/"MS"`). A row lag equals a time lag only when spacing is verified. Quarter-start labels represent whole quarters, not one day.
- `.loc["1992"]` selects a year; partial date slices include endpoints. `shift(h)` aligns earlier values; initial unavailable pairs are NaN.
- Trailing m-MA uses current/prior rows. Centered MA uses before/after observations and creates endpoint NaNs. Centered 3-MA example: $(443 + 410 + 420)/3 = 424.33$ at middle quarter. Centered windows use future values and leak in real-time use.

ADDITIVE DECOMPOSITION

- $y_t = T_t + S_t + R_t$ in original units. Fitted $\hat{y}_t = T_t + S_t$; $R_t = y_t - \hat{y}_t$. Exact reconstruction follows from remainder definition; it does not establish forecast accuracy or causes.
- Classical workflow: moving-average trend; detrend; average by season; center/repeat seasonal effects; remainder. `seasonal_decompose(...,model="additive",period=m)` uses centered trend and fixed mean-zero seasonal effects; even periods need special centering and lose endpoints.
- Limitations: fixed seasonality, complete-window loss, outlier influence. Full-history components are descriptive fits, not forecasts.
- STL = Seasonal and Trend decomposition using Loess. `STL(y,period=12,seasonal=13,robust=True)`: period=cycle length; odd seasonal window smooths across same-season occurrences; robust reweighting reduces impact of large remainders; endpoint support is less balanced. Flexible components are not uniquely true.
- Seasonal adjustment: $y_t^{adj} = y_t - S_t = T_t + R_t$; retains units/level and unusual departures while removing recurring pattern. It is *not* $y_t - T_t$ and not seasonal differencing. **HW5a audit lesson:** cement 2006 Q4→2007 Q1 observed change −311 thousand tonnes; classical adjusted −88.1; STL adjusted −0.6. Most apparent drop was seasonal; additive effects are constant amounts, not percentages.

LAGS + ACF

- Lag h compares y_t with y_{t-h}. Complete quarterly lag 4=same quarter prior year. Lag plot x=earlier, y=current. Seasonal ACF peaks often occur at multiples of period; opposite seasons may be negative; trend can create persistent positive ACF.
- Sample ACF:

$$r_h = \frac{\sum_{t=h+1}^{T}(y_t - \bar{y})(y_{t-h} - \bar{y})}{\sum_{t=1}^{T}(y_t - \bar{y})^2}$$

Uses one full-series mean/denominator, so differs slightly from Pearson correlation on shortened pairs. ACF measures linear dependence.

- White noise: mean 0, constant finite variance, zero ACF at nonzero lags; iid is sufficient, not necessary. Approx pointwise 95% bounds $\pm 1.96/\sqrt{T}$. Many inspected lags increase chance crossings. Inside bounds/zero ACF does not prove independence.
- Ljung–Box null through H: population autocorrelations at lags $1 \ldots H$ are jointly zero. `acorr_ljungbox(x,lags=[10])` gives one joint test through 10. Small p rejects; large p means compatible, not proof of independence/white noise. **HW5a dependence check:** quarterly cement change ending 2007 Q1=−311 (2006 Q4 to 2007 Q1); annual change=+113 (2006 Q1 2027 to 2007 Q1 2140). For annual-change series, Ljung–Box $Q(8) = 54.94$, $p = 4.54 \times 10^{-9}$: remaining autocorrelation. This does not itself test stationarity or prove more differencing is needed.

STATIONARITY + DIFFERENCING

- Stationary: joint distribution unchanged by time shift; changing mean, spread, or fixed seasonality violates it. Stationary series may remain dependent. Constant expectation is insufficient: random-walk level variance grows with time.
- Ordinary difference $\Delta y_t = y_t - y_{t-1}$ = change per observation interval; `diff()` loses first row. For random walk $y_t = y_{t-1} + e_t$, differencing recovers increments; if expected next increment is zero, current level is a simple forecast.
- Seasonal difference $\Delta_m y_t = y_t - y_{t-m}$; monthly annual `diff(12)` loses first 12. Seasonal adjustment subtracts estimated S_t and retains level; seasonal difference subtracts observed prior season and produces change.
- Log seasonal difference $\log y_t - \log y_{t-m} = \log(y_t/y_{t-m})$; .10 means ratio $e^{.10} = 1.105$, about 10.5% growth. Logs can stabilize proportional seasonal swings.
- Inspect time plot + ACF after transformation. Remaining ACF can be useful stationary dependence. Unnecessary differencing may introduce dependence. Second ordinary difference is change in first differences, not lag-2 difference.

LIGHT CODE INTERPRETATION + TRAPS

- `series.diff()` means previous row; `diff(4)` means same quarter last year only with complete quarterly spacing. `shift(h)` aligns levels without subtracting.
- `rolling(3,center=True).mean()` uses future row and produces endpoint NaNs. `dropna()` changes which observations enter calculations.
- `seasonal_decompose(y,model="additive",period=12)` returns `trend`, `seasonal`, `resid`. Seasonally adjusted is `y-result.seasonal`.
- A decomposition fitted on the complete series may describe it, but cannot be cited as out-of-sample predictive evidence.
- Always identify tested series: level, adjusted series, remainder, ordinary difference, or seasonal difference. The same ACF/test has different meaning on each. **Not Part A emphasis:** 12a forecasting is after the announcement's "concepts, theory, and principles through lecture 11b." Keep only leakage/forecast distinctions already necessary to interpret centered windows and historical decomposition; omit detailed forecasting models/metrics.

FINAL EXAM CHECKLIST

- Name observation/reference, features, units, scale, and time interval.
- Distinguish score from rule/flag/outcome; read score direction.
- Distinguish evidence-changing settings (k, features, scaler, reference) from threshold-only changes.
- For time series, verify calendar before interpreting row lags; state what differencing changes the quantity into.
- State null and tested lags for Ljung–Box; reject/compatible, never "prove independence."
- State what evidence does NOT prove: association≠causation; reconstruction≠forecast accuracy; anomaly≠fault; cluster≠population.